

Conditional Diffusion Models as Surrogate Samplers for the 2D Ising Model

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Abstract

We train a temperature-conditioned denoising diffusion probabilistic model to generate equilibrium configurations of the 2D Ising model on a 64×64 lattice. The model uses a UNet with adaptive group normalization, self-attention, and separate sinusoidal embeddings for the diffusion timestep and physics temperature, trained on Swendsen-Wang configurations across 51 temperatures spanning $T \in [1.0, 4.0]$. Energy and magnetization are reproduced to within statistical fluctuations across the full range; the spin-spin correlation function at T_c yields $\eta = 0.151 \pm 0.016$, compatible with the SW reference (0.125 ± 0.003) at $\sim 1.5\sigma$. Fluctuation-derived observables C_v and χ show a modest overshoot near T_c that diminishes with continued training. The scalar training MSE plateaus after the first epoch while observable-level errors continue falling by several orders of magnitude, establishing the training loss as a poor proxy for thermodynamic fidelity. Respaced sampling produces more decisively bimodal output than deterministic DDIM, eliminating a low-temperature C_v artifact. The model generalizes to unseen temperatures without degradation, and longer training improves fluctuation observables more than larger dataset size. We discuss why physics-informed loss terms face a structural mismatch with equilibrium statistical mechanics: unlike PDE-constrained systems, equilibrium configurations have no per-sample residual to penalize.

1. INTRODUCTION

Generating equilibrium configurations of statistical mechanical systems by Monte Carlo sampling is exact in principle but expensive in practice, particularly near phase transitions where autocorrelation times diverge. Generative models offer an alternative: once trained, they produce independent samples at fixed cost per configuration, regardless of proximity to criticality. This makes them attractive as surrogate samplers for systems where the Boltzmann distribution is known but costly to sample from directly.

The two-dimensional Ising model is a natural testbed for this idea. Its thermodynamics are exactly solvable in the infinite-size limit, its critical behavior at $T_c \approx 2.269$ is well characterized, and its binary spin structure on a square lattice is simple enough to isolate the performance of the generative model from complications of the target system.

Lee et al. [1] recently evaluated several generative architectures on this problem and found that diffusion models reproduce first-moment thermodynamic quantities (energy, magnetization) with high fidelity, but struggle with fluctuation-derived observables (specific heat C_v , susceptibility χ) near the critical temperature. They trained a model with two temperature granularities around T_c and identified a continuous-to-discrete mismatch, where Gaussian diffusion on binary spins introduces ambiguous sites that degrade ensemble statistics, as a key limitation.

In this work we replicate and extend their findings. We train a single temperature-conditioned denoising diffusion probabilistic model (DDPM) on a 64×64 Ising

lattice, following the UNet noise-prediction architecture introduced by Ho et al. [2] and incorporating subsequent improvements: a cosine noise schedule [3], adaptive group normalization for conditioning [4], self-attention at multiple resolutions [5], and deterministic DDIM sampling [6]. The physics temperature T enters through a separate embedding path from the diffusion timestep t , allowing a single model to generate configurations across the full temperature range $T \in [1.0, 4.0]$.

We pursue three goals. First, we want to characterize the model’s behavior in detail: how training loss relates to thermodynamic fidelity, how generated configurations compare to Swendsen-Wang ground truth across temperatures, and where the model succeeds or fails as a function of proximity to T_c . Second, we investigate whether architectural improvements can close the gap on fluctuation-derived observables (C_v , χ) and structural properties identified as limitations by Lee et al. Third, we examine a more fundamental question: whether physics-informed loss terms, which have shown promise in PDE-constrained settings [7], can be meaningfully applied to equilibrium statistical mechanics, where individual configurations have no “correct” energy at a given temperature and fluctuations are not noise but physical signal.

2. METHODS

2.1. Data generation

Training data was generated using the Swendsen-Wang (SW) cluster algorithm on a 64×64 square lattice. We chose SW over single-spin Metropolis updates because

cluster algorithms produce nearly uncorrelated configurations at every sweep. This eliminates the need for decorrelation spacing between samples and makes bulk generation substantially faster. Lee et al. used Metropolis sampling with decorrelation, which is viable but slower and requires careful calibration of the thinning interval at each temperature.

We sampled at 51 equally spaced temperatures $T \in [1.0, 4.0]$, with 1000 thermalization sweeps at each temperature. Two datasets were generated: one with 2000 and one with 5000 configurations per temperature, to test whether model performance is data-limited at this scale.

Data augmentation exploits the exact symmetry of the Ising model on a square lattice. Each configuration is randomly rotated by $k \times 90^\circ$ ($k \in \{0, 1, 2, 3\}$) and spin-reflected with probability $\frac{1}{2}$, applied on the fly in the DataLoader. This effectively multiplies the dataset by a factor of 8 without additional storage or introducing correlations within batches.

2.2. DDPM framework

Denoising diffusion probabilistic models (DDPMs) [2] define a generative process through two stages: a fixed forward process that gradually corrupts data into Gaussian noise, and a learned reverse process that recovers structured samples from noise.

Forward process. Given a data sample $\mathbf{x}_0$, the forward process defines a Markov chain that gradually adds Gaussian noise over T_{diff} steps. At each step, the sample is scaled down and noise is added according to a variance schedule $\{\beta_t\}_{t=1}^{T_{\text{diff}}}$, where $\beta_t \in (0, 1)$ controls how much noise is introduced at step t :

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}). \quad (1)$$

A key property is that $\mathbf{x}_t$ can be sampled directly from $\mathbf{x}_0$ at any t without iterating through intermediate steps. Defining $\alpha_t = 1 - \beta_t$ and the cumulative product $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, which tracks how much of the original signal remains at step t ,

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}). \quad (2)$$

Here $\sqrt{\bar{\alpha}_t}$ scales the original data and $(1 - \bar{\alpha}_t)$ is the variance of the accumulated noise. As $t \rightarrow T_{\text{diff}}$, $\bar{\alpha}_t \rightarrow 0$ and the sample converges to pure Gaussian noise. This closed-form expression is used during training to jump directly to a random noise level without simulating the chain. The schedule $\{\beta_t\}$ controls how quickly information is destroyed. Its effect is most naturally characterized through the signal-to-noise ratio $\text{SNR}(t) = \bar{\alpha}_t / (1 - \bar{\alpha}_t)$, which should decrease smoothly on a log scale to avoid forcing the model to learn abrupt denoising transitions at particular timesteps.

Reverse process and training objective. The reverse process approximates $q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ with a learned Gaussian:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \sigma_t^2 \mathbf{I}). \quad (3)$$

Ho et al. [2] showed that instead of predicting the mean $\boldsymbol{\mu}_\theta$ directly, the model can be reparameterized to predict the noise $\boldsymbol{\epsilon}$ that was added during the forward process. The training objective then reduces to a mean-squared error between the true and predicted noise:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} [\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2], \quad (4)$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the noise sampled during the forward process and $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ is the network’s prediction given the noised input and the timestep. This objective is a simplification of the variational lower bound (VLB), which is a principled bound on the negative log-likelihood of the data under the model. The full VLB decomposes into a sum of per-timestep KL divergences between the true denoising posteriors $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$ and the learned approximations $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$, and can be optimized directly if the model also predicts the reverse-process variance [3]. In practice, $\mathcal{L}_{\text{simple}}$ implicitly reweights these terms, down-weighting contributions from small t (where denoising is easy) and up-weighting large t , and despite lacking a formal probabilistic justification it produces higher sample quality than direct VLB optimization [2].

Sampling. Generation starts from pure Gaussian noise $\mathbf{x}_{T_{\text{diff}}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iterates the reverse process step by step, producing progressively cleaner samples until a final configuration $\mathbf{x}_0$ is obtained. In the original DDPM formulation this requires one model evaluation per timestep, typically 1000 in total, which is expensive.

Two approaches reduce this cost. Nichol and Dhariwal [3] showed that one can skip timesteps by sampling at, say, every 10th or 20th step, recomputing the transition distributions for the larger gaps. This is called respaced sampling and retains the stochastic character of the original process. Song et al. [6] proposed DDIM, which reformulates the reverse process so that it can be evaluated at any subset of timesteps without recomputing transition distributions. At $\eta = 0$, DDIM becomes fully deterministic: the same starting noise always produces the same output, which is useful for reproducibility. Both methods allow generation in 50 to 100 steps rather than the full 1000, with negligible quality loss for natural images. Whether this holds for the lower-entropy distributions of Ising configurations is an empirical question we address in Section 3.

Conditioning. An unconditional DDPM learns the marginal data distribution without distinguishing sub-populations. For our application the model must produce configurations appropriate to a specific physics temperature T . This is achieved by providing T as an additional input to the noise prediction network alongside the diffusion timestep t . The mechanism by which the conditioning signal enters the network is an architectural choice discussed in Section 2.3.

2.3. UNet architecture

The noise prediction network $\boldsymbol{\epsilon}_\theta$ is a UNet [8], an encoder-decoder architecture with skip connections originally

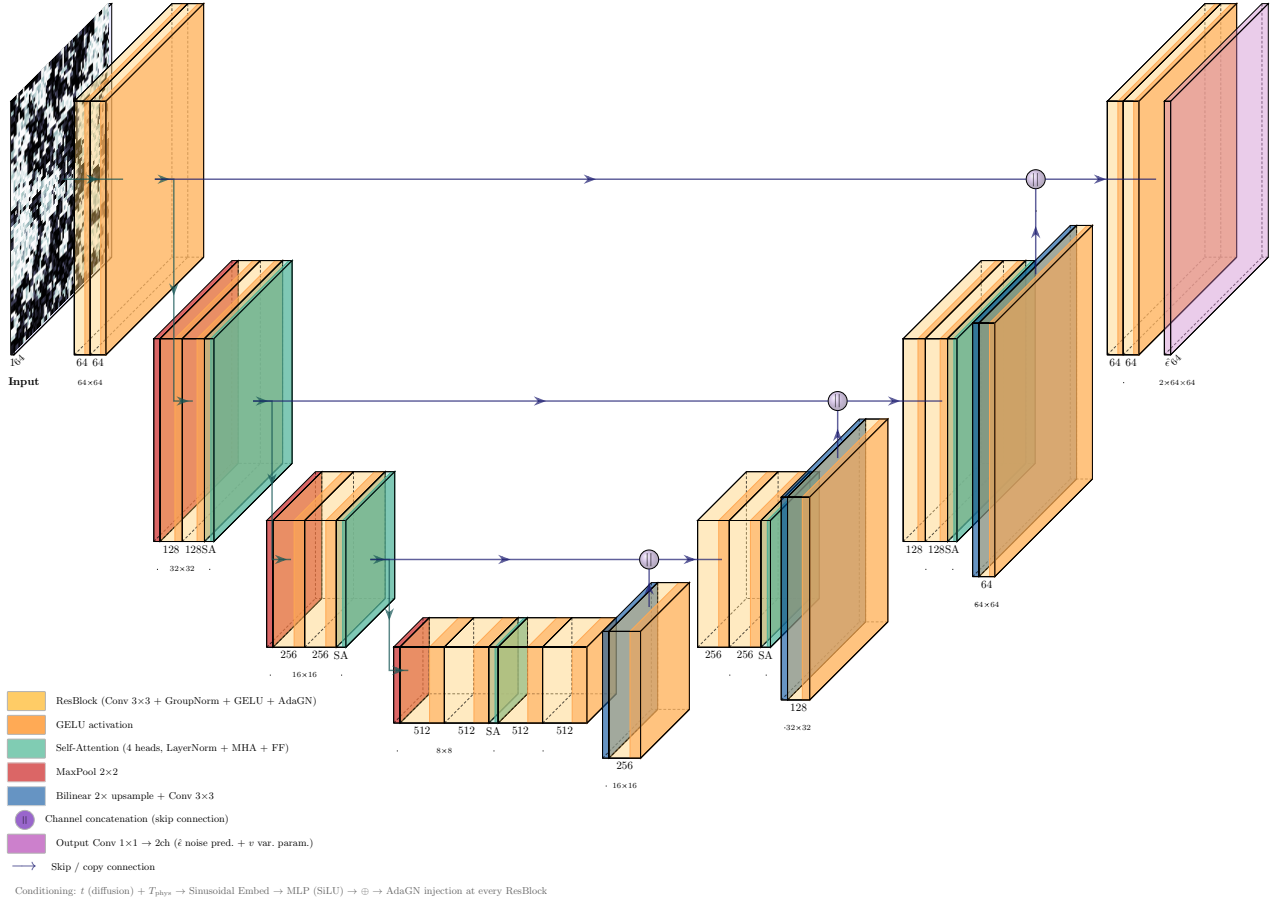


Figure 1: UNet Architecture

developed for medical image segmentation and first adapted to diffusion models by Ho et al. [2]. The encoder compresses the input through successive down-sampling stages, capturing features at multiple spatial scales. The decoder mirrors this structure in reverse, reconstructing the full-resolution output. Skip connections between corresponding encoder and decoder stages preserve fine-grained spatial detail that would otherwise be lost through the bottleneck. This multi-scale design is well suited to the Ising problem: local spin neighborhoods are captured at high resolution, while domain-scale structure and long-range correlations are represented in the lower-resolution feature maps.

Channel progression. The input is a single-channel 64×64 tensor (the noised spin configuration $\mathbf{x}_t$). The encoder maps this through channel dimensions $1 \rightarrow 64 \rightarrow 128 \rightarrow 256$, with max-pooling halving the spatial resolution at each stage. The bottleneck operates at 8×8 with 512 channels. The decoder mirrors the encoder back to 64 channels at 64×64 resolution. A final 1×1 convolution produces a 2-channel output: the predicted noise ϵ_θ and a variance parameter $\mathbf{v}$ used for learned-variance extensions of the loss (Section 2.2). The full architecture is shown in Figure 1; internal details of the building blocks are given in Appendix B.

Residual blocks. Each encoder and decoder stage consists of a residual block [9] containing two 3×3

convolutions with GELU activations and group normalization. A 1×1 convolution on the skip path matches channel dimensions where they change. Residual connections stabilize gradient flow through the deep network and allow each block to learn a correction to its input rather than a full transformation, which is particularly important given that the UNet is evaluated hundreds of times in sequence during generation.

Conditioning via adaptive group normalization.

The diffusion timestep t and the physics temperature T are each mapped to a 256-dimensional vector through separate sinusoidal embeddings [5] followed by a two-layer MLP with SiLU activations [2]. The two embedding vectors are summed to produce a single conditioning signal $\mathbf{e} \in \mathbb{R}^{256}$. This combined embedding enters each residual block through adaptive group normalization (AdaGN) [4]: after the second group normalization, a learned linear projection maps $\mathbf{e}$ to a per-channel scale and shift,

$$\text{AdaGN}(\mathbf{h}, \mathbf{e}) = (1 + \gamma(\mathbf{e})) \cdot \text{GroupNorm}(\mathbf{h}) + \beta(\mathbf{e}), \quad (5)$$

where γ and β are the scale and shift predicted from $\mathbf{e}$. This is applied after normalization but before the residual addition, so the conditioning modulates every feature map at every resolution. Summing the two embeddings rather than concatenating them keeps the conditioning dimension small while still allowing the model to represent joint effects of t and T . Crucially, the physics

temperature and the diffusion timestep share no parameters before the summation: their sinusoidal bases, MLP weights, and learned projections are entirely independent, enforcing the separation described in Section 2.2.

Self-attention. At the 32×32 , 16×16 , and 8×8 resolution stages, multi-head self-attention layers [5] with 4 heads are inserted after each residual block, in both the encoder and the decoder. Each spatial position is treated as a token with the channel dimension serving as the feature vector. Layer normalization is applied before the attention operation, and a feed-forward network (LayerNorm, linear, GELU, linear) follows with a residual connection. Self-attention allows the model to capture dependencies between distant lattice sites within a single forward pass, which is important near the critical temperature where the correlation length diverges and domain structure spans the full lattice.

Upsampling. The decoder uses bilinear interpolation followed by a 3×3 convolution at each upsampling stage, rather than transposed convolutions. This avoids the checkerboard artifacts that arise from the uneven overlap pattern of transposed convolution kernels [10].

2.4. Training

The training procedure follows established practices from the diffusion modeling literature, with a cosine noise schedule defining the forward process.

Noise schedule. We use the cosine schedule of Nichol and Dhariwal [3], which defines $\bar{\alpha}_t$ through a shifted cosine function:

$$\bar{\alpha}(t) = \frac{f(t)}{f(0)}, \quad f(t) = \cos\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right)^2, \quad (6)$$

with a small offset $s = 0.008$ that prevents the noise level from being too small near $t = 0$. Compared to the linear schedule used by Ho et al. [2], the cosine schedule distributes the signal-to-noise ratio more evenly on a log scale across timesteps, avoiding the problem of spending too many steps in noise regimes where little information is destroyed or recovered. The effect of the schedule is most naturally characterized through the log signal-to-noise ratio

$$\log \text{SNR}(t) = \log \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t} = 2 \log \cot\left(\frac{\pi}{2} \cdot \frac{t/T + s}{1 + s}\right), \quad (7)$$

which crosses zero at $t/T \approx 1/2$. We return to this quantity when decomposing the training loss by diffusion timestep in Section 3.1.

Exponential moving average. We maintain an exponential moving average (EMA) of the model parameters with a decay factor of 0.9999, following Nichol and Dhariwal [3]. At each training step, the shadow parameters are updated as $\hat{\theta} \leftarrow \gamma \hat{\theta} + (1 - \gamma)\theta$. The EMA weights are used for all evaluation and sampling. This smooths out the noise in individual gradient steps and

consistently produces better sample quality than the raw training weights, particularly for fluctuation-sensitive observables.

Learning rate warmup. The learning rate follows a linear ramp from 0 to the target value of 10^{-4} over the first 200 optimizer steps, then remains constant. Neither Ho et al. [2] nor Nichol and Dhariwal [3] used warmup in their original implementations, but our architecture includes self-attention layers at multiple resolutions, which produce sharper loss landscapes in the early stages of training. Warmup allows the Adam optimizer to accumulate accurate gradient moment statistics before taking full-sized steps, reducing the risk of instability at initialization [11].

Gradient clipping. Gradient norms are clipped to a maximum of 1.0 before each optimizer step. This guards against occasional large gradients that can arise from high-noise timesteps (large t) or from the attention layers, where the softmax can amplify gradient magnitudes through low-entropy attention distributions.

Training runs. We trained two models to assess whether performance is data-limited at this architecture size. Both use the same architecture described in Section 2.3 with identical hyperparameters. The first model (Ising DDPM) was trained for 100 epochs on the smaller dataset (51 temperatures $\times$ 2000 configurations per temperature). The second (Ising DDPM 2) was trained for 80 epochs on the larger dataset (51 temperatures $\times$ 5000 configurations per temperature). Training was performed on two different GPUs (NVIDIA RTX 5070 Ti and NVIDIA A100) with a batch size of 64 and 256 respectively and the Adam optimizer at a learning rate of 10^{-4} . All evaluation uses EMA weights and DDIM or respaced sampling with $\eta = 0$.

3. RESULTS

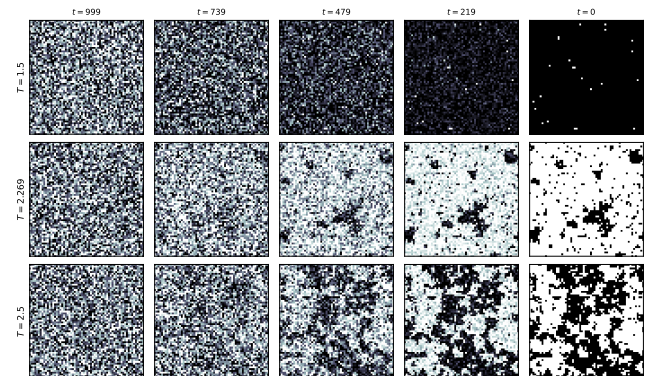


Figure 2: Denoising trajectory of the trained model at three physics temperatures. Starting from Gaussian noise, the reverse process produces ordered, critical, and disordered configurations depending on the conditioning temperature T .

Figure 2 shows representative denoising trajectories at temperatures below, near, and above T_c . The model

produces visually plausible configurations across the full temperature range: large aligned domains in the ordered phase, fractal-like structure near criticality, and disordered configurations at high temperature. We now assess quantitatively how well these samples reproduce the thermodynamic properties of the Ising model.

3.1. Training dynamics

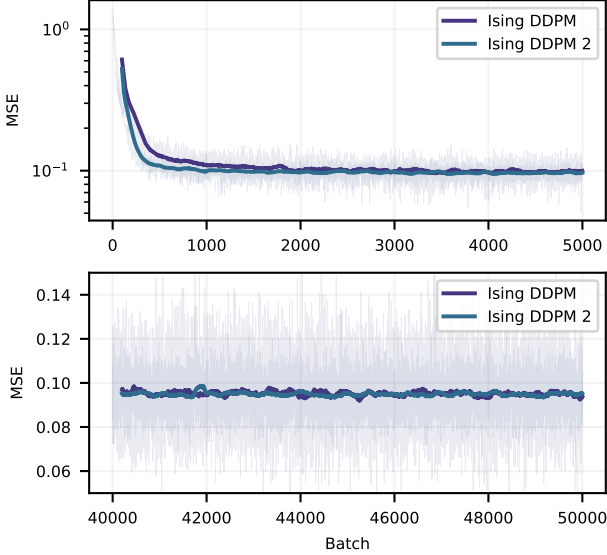


Figure 3: Training loss for the two models. Top: first 5000 batches, showing the initial rapid descent. Bottom: loss at a later training stage (around batch 40 000), where the MSE has long since plateaued and fluctuates noisily between 0 and 0.2.

The MSE loss drops sharply during the first few hundred batches and plateaus within the first epoch (Figure 3). After this initial descent, the loss fluctuates without a clear downward trend for the remainder of training. Taken at face value, this would suggest that the model has converged and further training is unproductive. As we show in Section 3.2, this impression is misleading: thermodynamic observables continue improving substantially through later epochs.

The temperature-resolved loss (Figure 4) reveals why the aggregate MSE is a poor diagnostic. The loss is lowest in the ordered phase, where configurations are highly structured and the noise prediction task is comparatively easy. It rises steadily through the disordered phase, not because the model performs worse there, but because the space of valid configurations is much larger: at high temperature, many distinct spin arrangements are thermodynamically equivalent, and the model cannot (and should not) collapse onto a single prediction. A low MSE in the disordered phase would indicate memorization of training samples rather than correct sampling from a broad equilibrium distribution.

The loss decomposed by diffusion timestep (Figure 5) shows a clear peak at intermediate t values where the signal-to-noise ratio passes through unity. This is expected: at small t the input is nearly clean and denoising is trivial, at large t the input is nearly pure noise and the

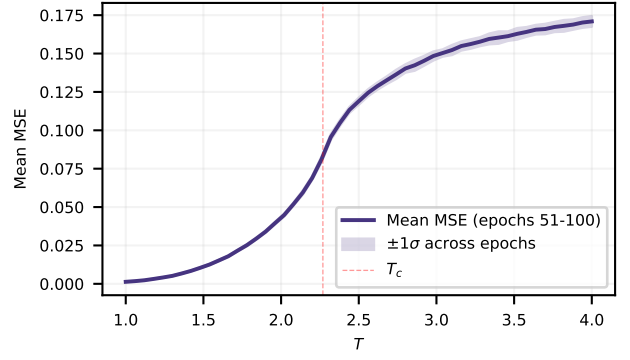


Figure 4: MSE loss averaged over late epochs as a function of physics temperature. The loss is lowest in the ordered phase ($T < T_c$) and rises steadily into the disordered phase.

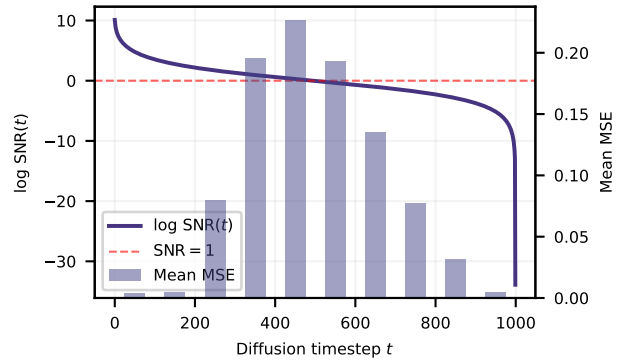


Figure 5: MSE loss by diffusion timestep bucket, with the signal-to-noise ratio $\text{SNR}(t)$ overlaid Eq. 7. The peak loss occurs near $\text{SNR} \approx 1$, where the model must distinguish physical signal from added Gaussian noise.

optimal prediction approaches zero, but near $\text{SNR} \approx 1$ the model faces the hardest discrimination task between physical structure and Gaussian noise.

Together, these diagnostics establish that the MSE loss is too coarse-grained to track generative quality for this problem. We return to this point in Section 4.2.

3.2. Thermodynamic observables

Figure 6 shows the primary comparison between Ising DDPM and SW generated observables at the final training epoch. The first-moment quantities E and $|M|$ are reproduced with high fidelity across the full temperature range, with generated and SW curves nearly indistinguishable. This is consistent with the findings of Lee et al. [1], who observed similarly close agreement in first moments.

The peak specific heat reaches $C_v \approx 3.8$ versus SW ≈ 2.2 , and susceptibility $\chi \approx 95$ versus SW ≈ 70 . Since our SW reference already crosses between magnetization sectors efficiently, this overshoot cannot be attributed to ensemble independence between generated and reference samples. (Lee et al. [1] used Metropolis sampling, where inter-sector mixing is exponentially suppressed, so their reported C_v discrepancy likely has a larger ensemble-

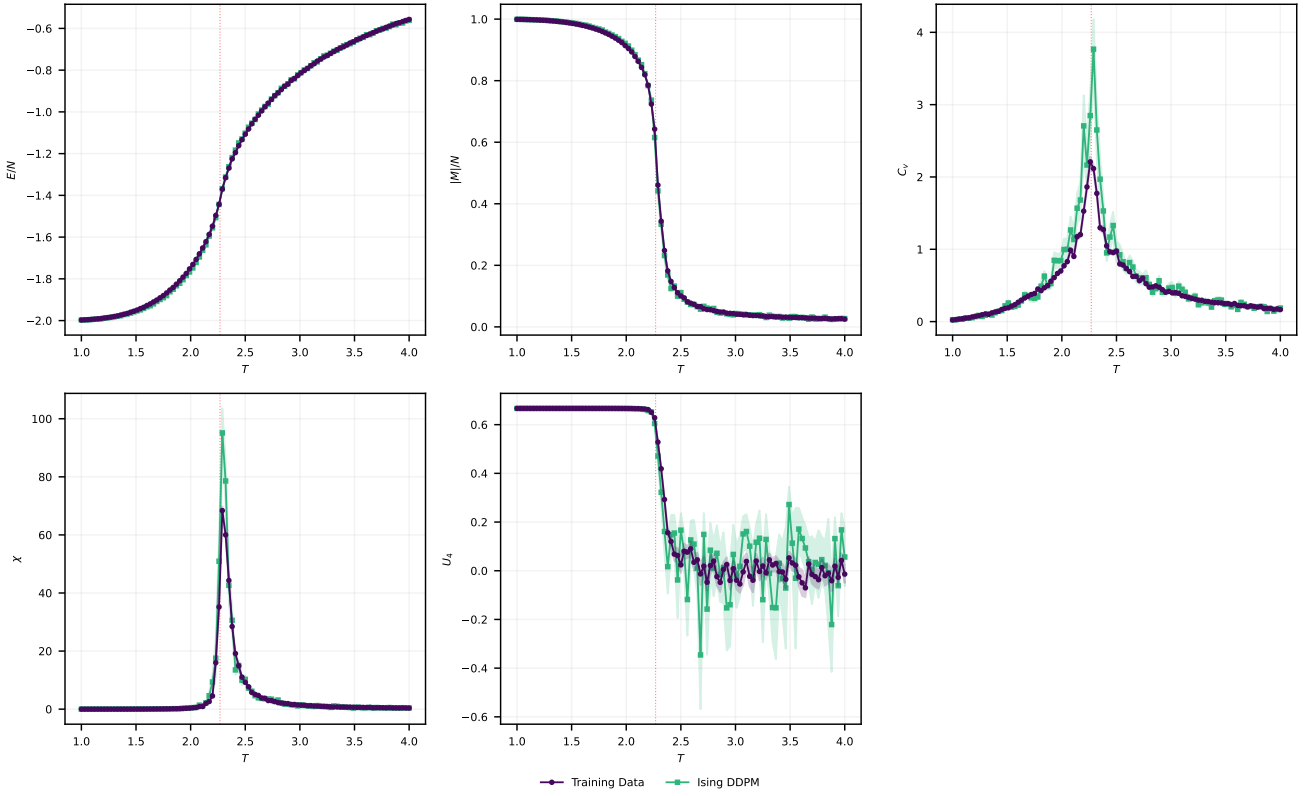


Figure 6: Thermodynamic observables as a function of temperature for the ground truth and the trained model (epoch 100, respaced sampling with 100 steps, 101 temperatures). From left to right, top to bottom: energy per spin E , absolute magnetization $|M|$, specific heat C_v , magnetic susceptibility χ and Binder cumulant U_4 . Error bands show bootstrapped 95% confidence intervals.

independence component.) The overshoot in our case reflects incomplete learning of the fluctuation structure at the longest correlation lengths near T_c , consistent with the continued improvement of C_v and χ through late training epochs (Figure 7). Since $C_v \propto \text{Var}(E)/T^2$ and $\chi \propto \text{Var}(M)/T$, overestimation indicates that the generated ensemble has slightly too much spread in its energy and magnetization distributions. The Binder cumulant U_4 , which depends on the ratio $\langle m^4 \rangle / \langle m^2 \rangle^2$ and is therefore insensitive to a uniform broadening of the magnetization distribution, does not show a corresponding overshoot. This pattern is consistent with the model having not yet fully converged in the critical region, where the diverging correlation length requires the network to correctly reproduce structure at all spatial scales simultaneously. Small residual imprecision in the longest-range correlations would broaden the energy and magnetization distributions without producing visible artifacts in individual configurations or shifting their means. The continued improvement at later epochs supports this interpretation: the critical region is simply where learning is slowest.

The Binder cumulant U_4 matches the SW values in the ordered phase and through the transition, but shows larger fluctuations in the disordered phase. This is not a systematic bias: above T_c the magnetization fluctuates around zero, and $U_4 = 1 - \langle m^4 \rangle / (3\langle m^2 \rangle^2)$ becomes sensitive to higher moments of a distribution centered at zero. The SW ground truth itself fluctuates in this regime, and the model reproduces that behavior with

somewhat larger swings.

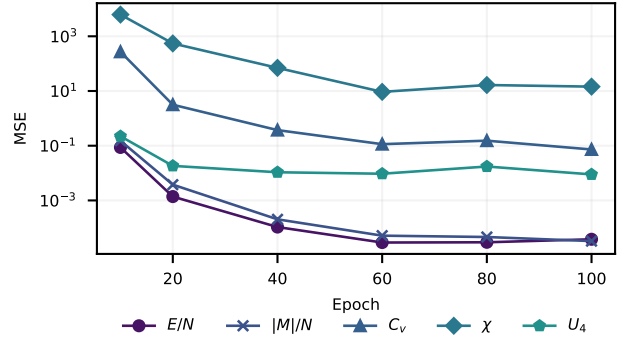


Figure 7: Mean squared error between generated and SW ground truth observables as a function of training epoch (respaced sampling, 100 steps, 100 samples per temperature).

The epoch progression (Figure 7) resolves the apparent contradiction between the stagnant training loss and the quality of the final results. The observable-level MSE decreases consistently from epoch 10 through epoch 100 for all quantities, even though the noise-prediction loss has long since plateaued. Energy and magnetization MSE drop by roughly four orders of magnitude (from $\sim 10^1$ to $\sim 10^{-3}$) between epochs 10 and 60, while C_v and χ MSE continue falling through epoch 100. First-moment observables (E , $|M|$) are already well matched

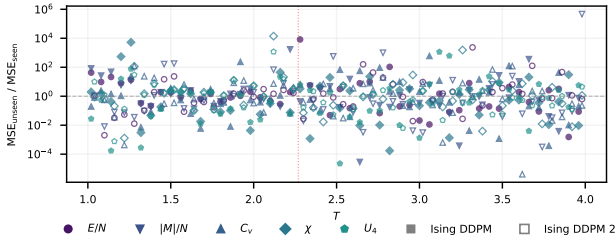


Figure 8: Ratio of the per-temperature MSE at unseen (interpolated) temperatures to the MSE at the nearest seen (training) temperature, shown for all observables and both models. The dashed line marks parity ($\text{MSE}_{\text{unseen}}/\text{MSE}_{\text{seen}} = 1$).

at early epochs and continue improving modestly. The second-moment quantities (C_v , χ) start with larger errors but show the steepest improvement, confirming that the model continues learning the variance structure of the equilibrium distribution throughout training. The Binder cumulant sits between the two groups, consistent with its dependence on higher moments that are easier to match than C_v and χ but harder than the means.

Figure 8 tests whether the temperature conditioning memorizes the 51 training temperatures or interpolates smoothly between them. The ratio $\text{MSE}_{\text{unseen}}/\text{MSE}_{\text{seen}}$ scatters symmetrically around unity across the full temperature range, with no systematic bias toward higher error at unseen temperatures. This holds for all five observables and for both models. The spread is large on a per-temperature basis (spanning several orders of magnitude), but this reflects the noise inherent in comparing single-temperature MSE values computed from 100 samples, not a generalization failure. The absence of any trend near T_c is notable: the model does not struggle more to interpolate in the critical region than elsewhere, even though the absolute errors are largest there.

The two models (Ising DDPM and Ising DDPM 2) show indistinguishable generalization behavior, reinforcing the conclusion that doubling the dataset size from 2000 to 5000 samples per temperature does not measurably improve performance. The longer-trained model in fact outperforms the larger-dataset model on fluctuation observables (C_v , χ), suggesting that additional training epochs matter more than dataset size at this architecture scale.

3.3. Sampling and binarization

The DDPM generates continuous-valued outputs that must be binarized to ± 1 spins. Sites where the raw output x_0 falls close to zero are ambiguous: they lie in a “gray zone” where the sign threshold could go either way. Lee et al. [1] identified this continuous-to-discrete mismatch as a source of error in fluctuation-derived observables.

Figure 9 (top panel) shows that the mean gray-zone fraction drops by roughly three orders of magnitude between epoch 10 and epoch 40, then plateaus near 10^{-6} . This indicates that as training progresses, the model

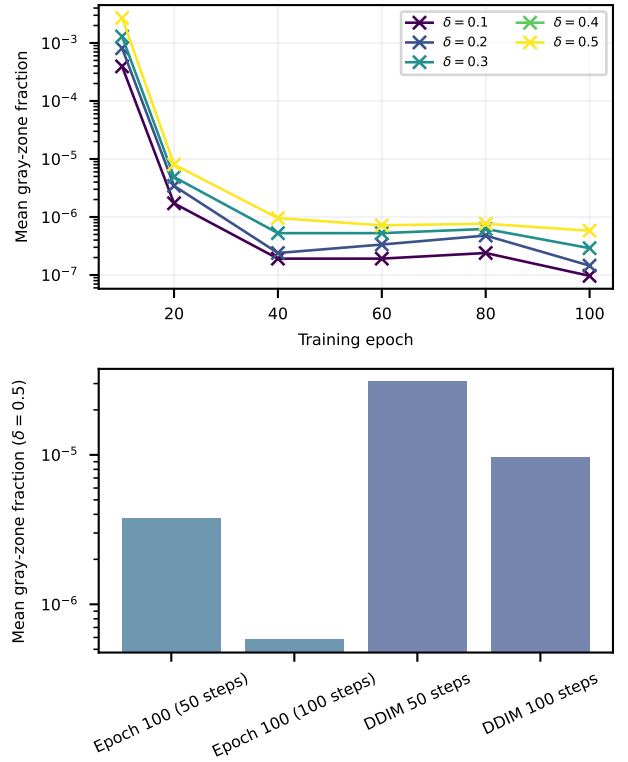


Figure 9: Mean gray-zone fraction (sites with $|x_0| < 0.5$) averaged over all temperatures. Top: evolution across training epochs under respaced sampling (100 steps). Bottom: comparison of respaced and DDIM sampling at epoch 100 with 50 and 100 steps.

learns to produce outputs more decisively pushed toward ± 1 . At the final epoch, fewer than one site per configuration on average falls in the ambiguous range, making binarization effectively a non-issue for downstream observables.

The bottom panel compares sampling strategies at epoch 100. Respaced sampling with 100 steps produces gray-zone fractions of order 10^{-6} , while DDIM at the same step count yields $\sim 10^{-5}$. Although both values are small in absolute terms, the difference has observable consequences: DDIM produces a spurious C_v spike of order 20–25 near $T^* \approx 1.35$ that does not appear in respaced output. The deterministic DDIM reverse path compounds $\hat{x}_0$ prediction errors across steps, while respaced sampling injects Gaussian noise at each iteration, reducing error accumulation and producing more decisively bimodal output. We used respaced sampling with 100 steps for all observable evaluations reported in this work.

3.4. Critical exponent

A stronger test than matching bulk averages is whether the model captures the spatial structure of the equilibrium ensemble. We probe this through the radially averaged spin-spin correlation function $G(r) = \langle s_i s_{i+r} \rangle - \langle s_i \rangle \langle s_{i+r} \rangle$, computed per sample via the Fourier power spectrum. At criticality $G(r)$ follows a power law $G(r) \sim r^{-\eta}$, where $\eta = 1/4$ in the thermodynamic limit. We fit η from the log-log slope of

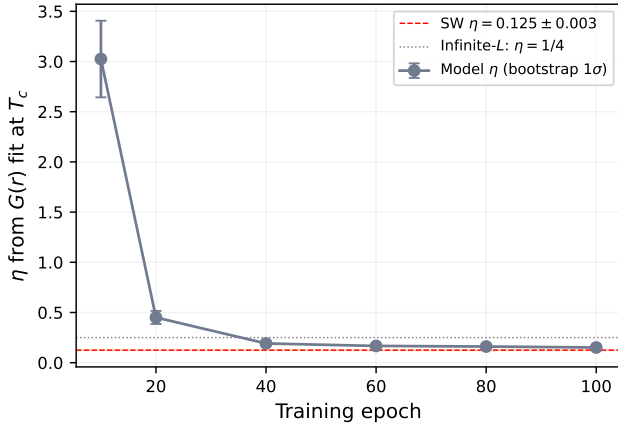


Figure 10: Critical exponent η at $c = 2.26$ from a log-log fit of $\bar{G}_r$ over $r \in [2, L/2]$, for each checkpoint. Error bars are 1σ from a sample-resampling bootstrap ($B = 400$). The model converges to $\eta_{100} = 0.151 \pm 0.018$, compatible with the SW reference ($\eta_{\text{SW}} = 0.125 \pm 0.003$, shaded band) at $\sim 1.5\sigma$. Dotted: $\eta = 1/4$ (infinite L).

$\bar{G}_r$ over $r \in [2, L/2]$ at $c = 2.26$ for each saved checkpoint, estimating uncertainties with a sample-resampling bootstrap.

Figure 10 tracks η across training. By the final checkpoint the model yields $\eta_{100} = 0.151 \pm 0.016$, within $\sim 1.5\sigma$ of the SW reference (0.125 ± 0.003). Both values sit below the thermodynamic-limit $\eta = 1/4$, which is the expected finite-size effect at $L = 64$. The exponent is still drifting downward at epoch 100, in line with the ongoing improvement of fluctuation observables (Figure 7).

4. DISCUSSION

4.1. Physics-informed diffusion and the equilibrium mismatch

The DDPM reproduces Ising thermodynamics using a purely data-driven objective: the model sees spin configurations labeled by temperature and learns to denoise them, with no explicit reference to the Hamiltonian or the Boltzmann distribution. A natural question is whether injecting physical knowledge into the training process could close the remaining gap near T_c .

For the Ising model, both the Hamiltonian $H[\mathbf{s}] = -J \sum_{\langle i,j \rangle} s_i s_j$ and the equilibrium distribution $P(\mathbf{s}) \propto \exp(-H[\mathbf{s}]/T)$ are known exactly. Other generative frameworks can exploit this. Normalizing flows, for instance, provide tractable likelihoods that in principle allow training directly against the Boltzmann distribution without any SW training data [12]. DDPMs have no such route. The reverse process defines an implicit distribution whose density cannot be evaluated, so there is no way to compare a generated sample’s likelihood against the known target during training.

The alternative is to add physics-informed terms to the loss. This has been demonstrated for PDE-constrained systems [7], where each generated sample either satisfies the governing equations or not, providing a clear

per-sample error signal. A PDE-constrained system provides a per-sample residual: each generated configuration either satisfies the governing equations or not. In equilibrium statistical mechanics, a configuration’s energy at temperature T is a random variable drawn from a broad distribution, and the fluctuations are the physics, not noise to be penalized.

This is a structural limitation of the DDPM framework for problems where the target distribution is known, not a tuning problem. In practice, the data-driven approach works well when training data is cheap to generate, as it is for the standard Ising model. The limitation becomes more consequential for systems where sampling is itself the bottleneck.

4.2. What the loss doesn’t tell you

The disconnect between the MSE loss and thermodynamic fidelity, documented in Sections 3.1 and 3.2, has practical consequences for how diffusion models should be evaluated in physics applications.

The MSE loss measures how well the model predicts the noise added at each diffusion timestep, averaged over all timesteps, temperatures, and spatial positions. It is a per-pixel, per-step quantity that knows nothing about the collective properties of the generated ensemble. Two models with identical MSE could produce very different thermodynamic statistics if their errors are distributed differently across temperatures, timestep ranges, or spatial scales. As shown in Section 3.2, the training MSE plateaus after the first epoch while observable-level errors continue falling by several orders of magnitude through epoch 100. A practitioner who relied on loss convergence for early stopping would discard a model that had not yet learned the physics it would eventually capture.

This suggests that for physics applications, the training loop should include periodic evaluation against the target observables, not as a loss term (which faces the difficulties discussed in Section 4.1) but as a diagnostic. Generating a modest number of samples at a few representative temperatures and computing E , $|M|$, C_v , and χ every few epochs is cheap relative to the training cost and provides a far more informative convergence criterion than the MSE.

5. CONCLUSION & FUTURE WORK

We trained a temperature-conditioned DDPM on the 2D Ising model and evaluated its ability to reproduce equilibrium thermodynamics across the full temperature range $T \in [1.0, 4.0]$. The model reproduces first-moment quantities (E , $|M|$) with high fidelity, and recovers the power-law spin-spin correlation at T_c with $\eta_{100} = 0.151 \pm 0.018$, compatible with the SW reference at $\sim 1.5\sigma$. Fluctuation-derived observables (C_v , χ) show a mild systematic overshoot near the critical temperature that diminishes with continued training, consistent with the critical region being the slowest to converge. The Binder cumulant U_4 matches the SW ground truth through the transition and fluctuates in the disordered phase where the quantity is inherently

noisy. Temperature conditioning generalizes smoothly to unseen temperatures, and doubling the dataset size from 2000 to 5000 samples per temperature produced no measurable improvement, suggesting that at this architecture size the model capacity rather than the training data is the limiting factor. Lee et al. [1] trained separate unconditional models, one on a coarse temperature grid and one on a finer grid concentrated around T_c ; our single conditioned model avoids this splitting entirely while achieving tighter agreement on the second-moment observables C_v and χ . The remaining deviations near T_c are better explained by the inherent difficulty of learning critical fluctuations than by binarization artifacts.

Lee et al. identified the continuous-to-discrete mismatch as a primary source of error in fluctuation-derived observables. Our analysis qualifies this conclusion: the gray-zone fraction is negligible (on the order of 1–2 ambiguous sites per 64×64 lattice) and depends more on the choice of sampling strategy than on the model itself, with respaced sampling producing more decisive outputs than DDIM at the same number of steps (Figure 9).

Several directions could extend this work. Alternative noise schedules, such as Laplace-weighted SNR schedules or learned schedules, may improve the allocation of model capacity across the temperature range. Extended loss formulations, in particular the variational lower bound with a learned variance [3], offer a more principled training objective than pure MSE and warrant further investigation. Finally, training on multiple lattice sizes simultaneously would test whether the model can learn finite-size scaling, though this would require an architecture capable of handling variable input dimensions.

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A. CONTRIBUTIONS

The training data pipeline builds on a joint Swendsen-Wang implementation from Exercise 3. The UNet architecture was designed collaboratively. Diffusion infrastructure, including noise schedules and sampling methods, was developed independently by each author and merged based on model performance. Analysis was discussed jointly but carried out separately.

B. ARCHITECTURE DETAILS

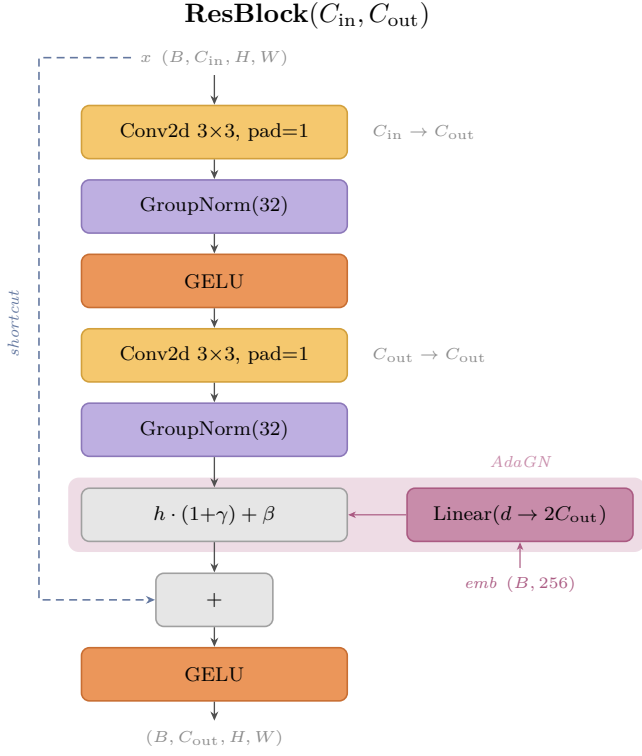


Figure 11: Residual Block Architecture

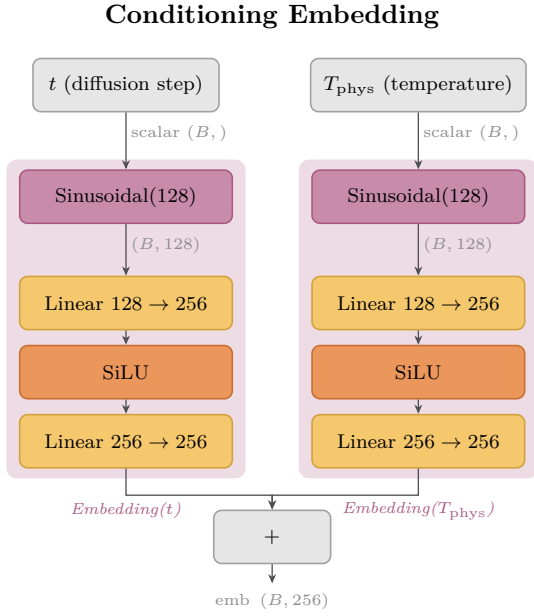


Figure 12: Embedding Block Architecture

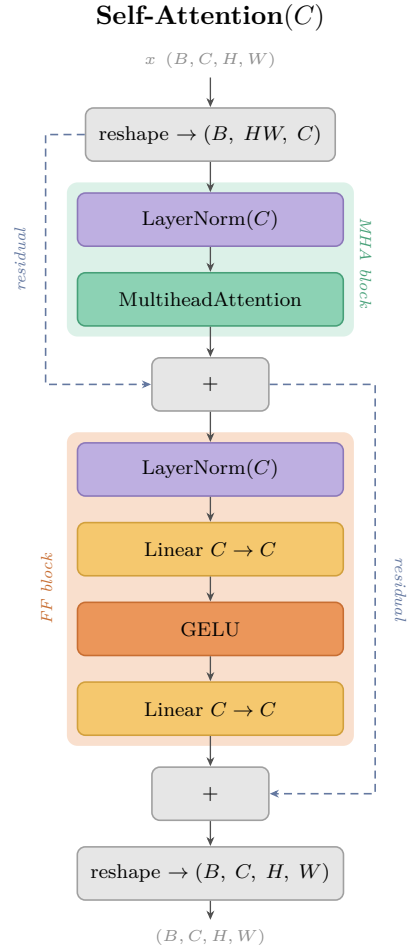


Figure 13: Self-Attention Block Architecture